ALGOMANIACS

TRIES

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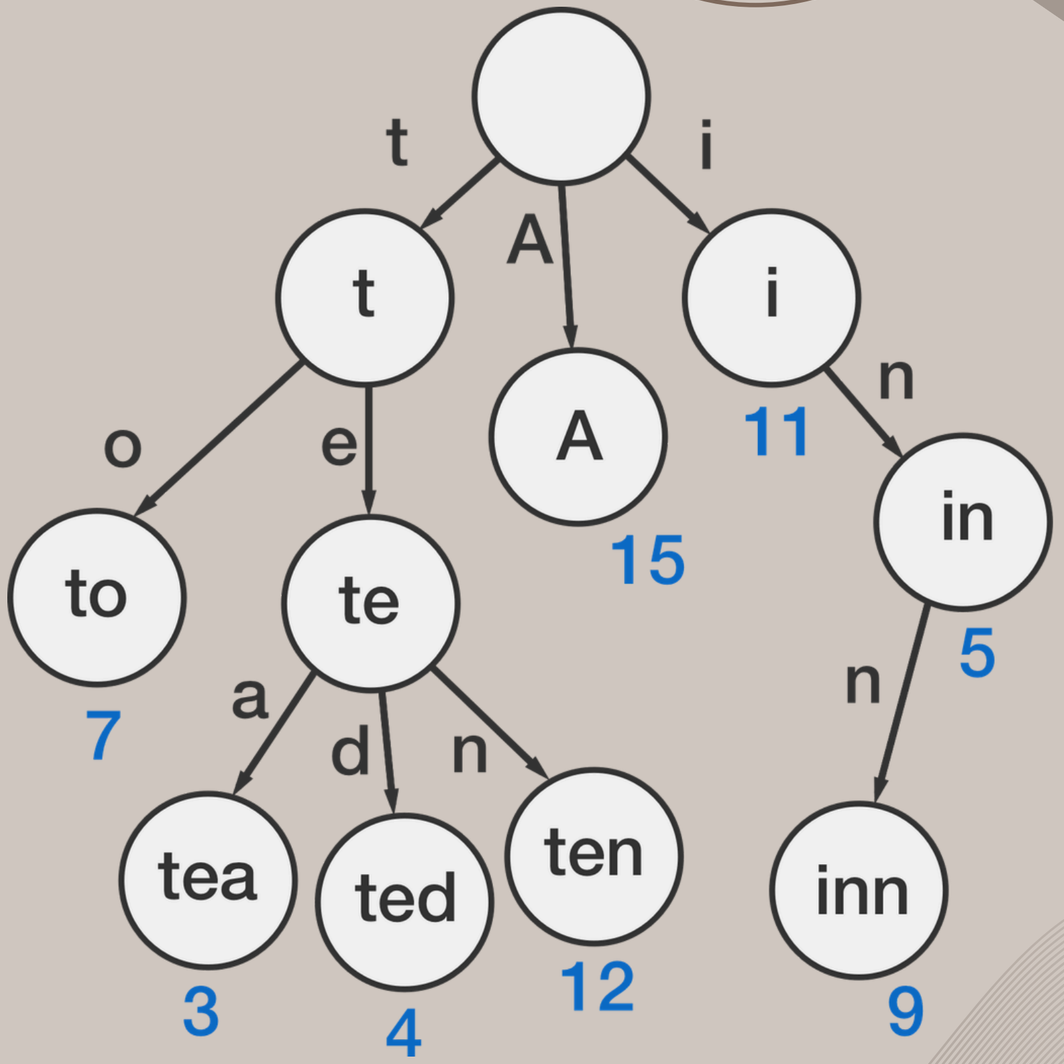


WHAT ARE TRIES?

Tries are a tree-like data structure containing nodes and edges.
Each Trie has an **empty root node**, with links to other nodes.
Each node of a Trie represents a string given by the path from the root to the node.
Each edge represents a character from a **fixed** alphabet.
Every node consists of a hashmap or an array of pointers to other nodes, with each index representing a character and some information.
A fun visualizer - link

IMPLEMENTATION

A vertex struct/array is all we need to dynamically make tries. When we encounter a vertex without an edge we want, we simply add a new vertex to our Trie and point our current vertex to the new vertex. Also if we want a static Trie, we can make a fixed array Trie and keep a count of number of nodes we have currently used
Time & Space complexity - O(K*(length))

```
const int K = 26;

struct Vertex{
    int next[K]; // next node
    bool output = false;

    Vertex(){ // constructor
        fill(begin(next), end(next), -1);
    }
};

vector<Vertex> t(1);

void add_string(string const& s){
    int v = 0;
    for(char ch : s){
        int c = ch - 'a';
        if(t[v].next[c] == -1){ // if no node, create new
            t[v].next[c] = t.size();
            t.emplace_back();
        }
        v = t[v].next[c];
    }
    t[v].output = true;
}
```

PROBLEMS

- Given a set of strings, do we have a string s in the set?
- Given a set of strings, how many strings have a prefix p in them?
- Given a set of strings, how many unique suffixes are there?
- Given a set of strings, find the kth lexicographically smallest string.
- https://codeforces.com/contest/455/problem/B - Subproblem: Given a group of *n* non-empty strings. During the game two players build the word together, initially the word is empty. The players move in turns. On his step player must add a single letter in the end of the word, the resulting word must be prefix of at least one string from the group. A player loses if he cannot move.

BIT TRIES

How do we use tries for numbers instead? Tries are powerful for prefix related questions as you saw in the previous question. What are the prefixes here?

Can we think a number in binary representation as a string? Is it helpful? Where do we start from MSB to LSB or from LSB to MSB?

PROBLEMS

- Given an array of integers, find two elements whose XOR is maximum.
- Given an array of integers, find the subarray with maximum XOR.
- Given an array of positive integers you have to print the number of subarrays whose XOR is less than K.
- https://atcoder.jp/contests/arc122/tasks/arc122_d

PERSISTENT TRIES

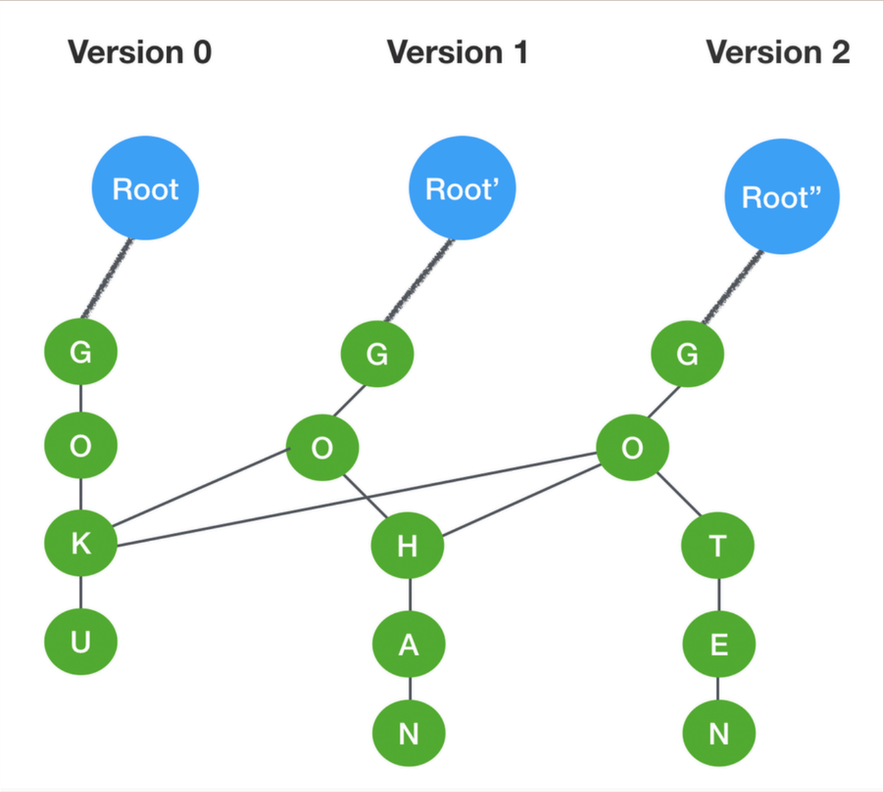
They store the history of strings. They work by creating a new root node and new nodes for each word in the string. It meanwhile adds all the unchanged nodes as they are, which still gives us a time and space complexity of $O(K*length_{string})$

```
const int K = 26;
const int MAXNODES = 7000000;

int nxt[MAXNODES][K];
bool output[MAXNODES];
int node_cnt = 1;

int add_string(int prev, const string& s){ // prev is previous tree
    int root = node_cnt++; // new node
    int cur = root;
    for(char ch : s){
        int c = ch - 'a';
        for(int i=0;i<K;i++){
            nxt[cur][i] = nxt[prev][i];
        }
        output[cur] = output[prev];
        nxt[cur][c] = node_cnt++; // new node
        cur = nxt[cur][c];
        prev = nxt[prev][c];
    }
    for(int i=0;i<K;i++){
        nxt[cur][i] = nxt[prev][i];
    }
    output[cur] = true;
    return root;
}
```

PERSISTENT TRIES

Let's look at a graph with three words: goku, gohan and goten

If we are too worried about memory overflow, we can use a hashmap to store the next node when needed instead of allocating K memory everytime

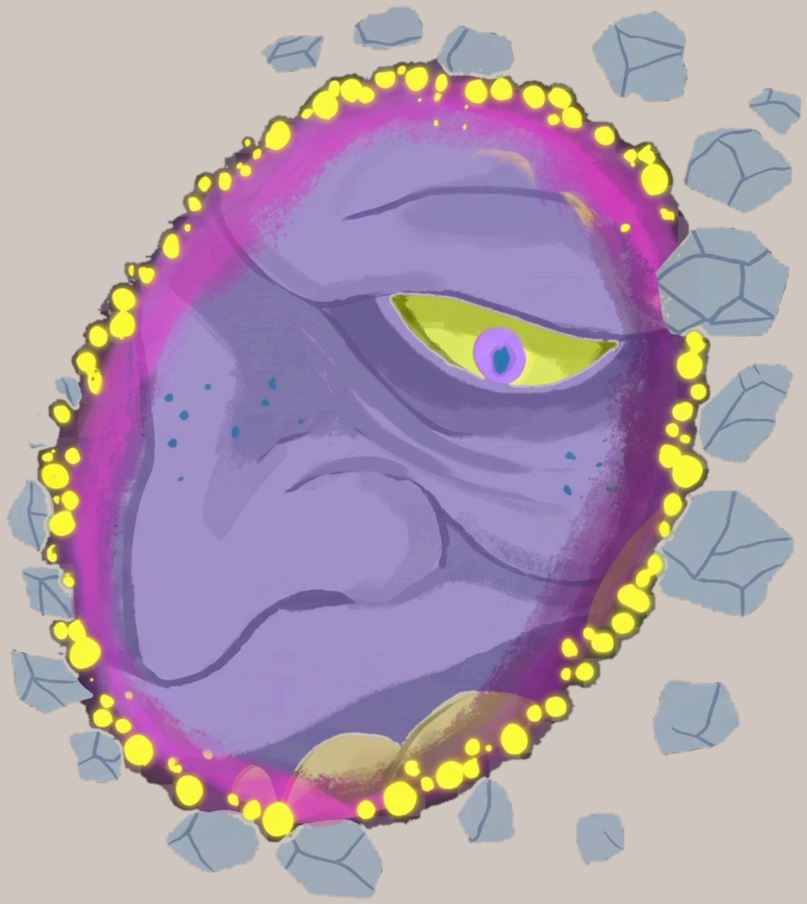


PROBLEMS

- Given an array of positive integers and Q queries of 2 types
  a. add a new number to the array
  b. find an index i in [L, R] such that A[i] xor X is maximized.
- Given a tree of N nodes each having a value V_u and Q queries of type (u,v,X) asking to find a node w belonging to the simple path between u and v such that V_w xor X is maximized.

MONSTER PROBLEM

https://www.codechef.com/ICPCAWOS25/problems/MAXFREQMEX



FURTHER USE

Single chains with multiple letters can be collapsed into an edge to save memory (Patricia Tries)
Tries can also be effectively used in finite automations such as in Aho-Corasick algorithm (requires KMP) and in suffix automations.

THANK YOU